Gaussian Process Latent Variable Models for Simplifying Image Data

Joey Hotz

University of Toronto, Department of Statistical Sciences

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Team Information: All aspects of this project (coding, writing the report, creating slides, project motivation, etc.) were done by Joey Hotz.

All code is available on GitHub: https://github.com/hotz1/STA2201-Final-Project

1 Introduction

The overarching goal of this project is to investigate the properties of Gaussian Process Latent Variable Models (GP-LVMs), with a focus on the application of these models to dimension reduction in a high-dimensional data setting.

My personal motivation for choosing this topic for my report is that the GP-LVM method is an

2 Methodology

In order to develop a full mathematical understanding of the GP-LVM methodology and how this method can be used for dimensionality reduction, we must first discuss two pre-requisite ideas which are crucial in motivating the GP-LVM.

2.1 Probabilistic PCA

2.2 Gaussian Processes

A Gaussian Process $\{X_t\}_{t \in T}$ is a stochastic process of random variables $\{X_t\}$ for t in some pre-determined indexing set T which has the property that the joint distribution of any any *finite* collection of random variables from this process will be a multivariate Gaussian distribution. [3]

In more formal mathematical notation, for any collection of distinct indices $\{t_1, \ldots, t_N\} \subset T$, the N-dimensional random vector $\begin{pmatrix} X_{t_1} & \cdots & X_{t_N} \end{pmatrix}^\mathsf{T} \in \mathbb{R}^N$ follows a multivariate Gaussian distribution on $\mathbb{R}^N$. Note that by this definition, every individual random variable X_t must follow a univariate Gaussian distribution.

Gaussian Processes are commonly used in regression settings as a 'surrogate model' for modelling unknown functions, as the framework for a Gaussian Process allows for a lot of flexibility wen learning an unknown function.

From a statistical perspective, a Gaussian Process defined on the indexing set T is entirely specified by two functions; a *mean function* $\mu : T \to \mathbb{R}$ which represents the expectation of the random variables, and a *covariance kernel function* $k : T \times T \to \mathbb{R}$, which represents the covariance between the random variables [3]. In most Gaussian Process regression settings,

2.3 Gaussian Process Latent Variable Models

Now that we have covered the fundamentals of PPCA and Gaussian Processes, we may turn our attention towards the primary methodological focus of this report.

3 Data Description

The primary image dataset which we use for our simulations involving this method is the `Fashion-MNIST` dataset [4], which is a publicly-available benchmarking dataset used in a variety of machine learning tasks. The data is

The primary dataset of interest for our simulation study is the `Fashion-MNIST` dataset [4], which is a popular dataset for machine learning benchmarking. The choice of this dataset is primarily motivated by the simulations in [1], which uses the 'original' MNIST dataset containing handwritten digits.

The Fashion-MNIST dataset is built into many common machine learning software libraries by default, including `PyTorch` [2].

Figure 3.1: One example from each of the ten classes of clothing items in the `Fashion-MNIST` dataset

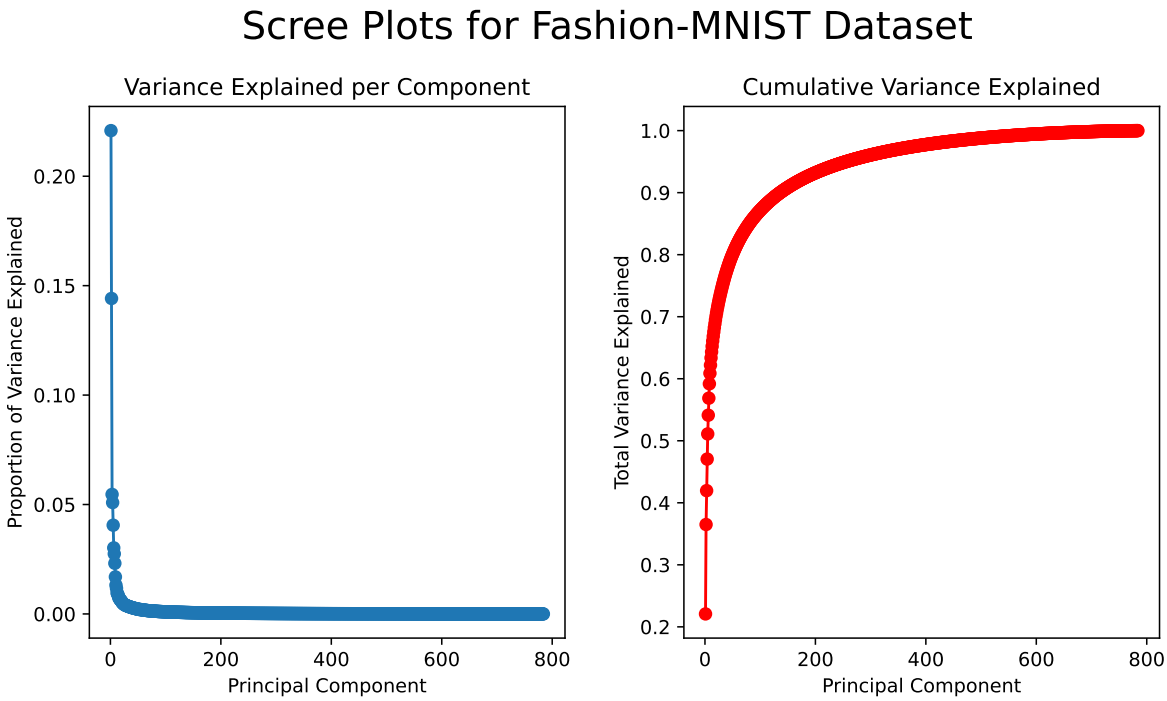
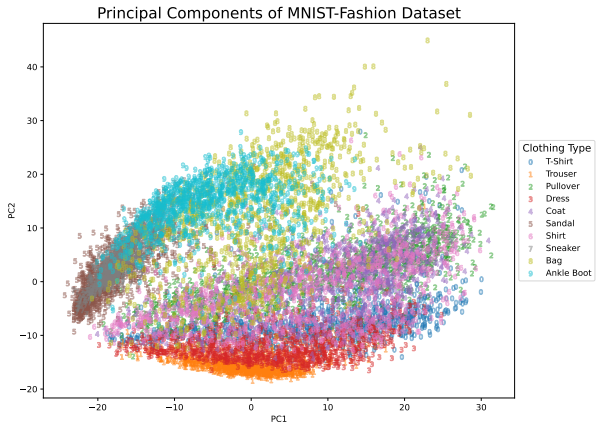
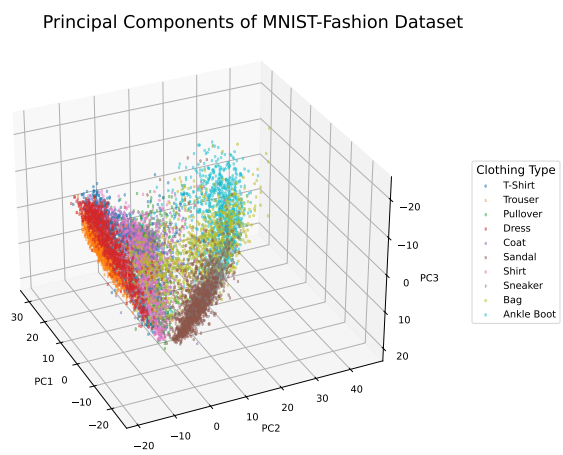


Figure 3.2: Scree plots for the principal components of the training data in the `Fashion-MNIST` dataset



(a) Top two principal components of clothing items in the `Fashion-MNIST` dataset



(b) Top three principal components of clothing items in the `Fashion-MNIST` dataset

Figure 3.3: Top principal components of clothing items, coloured by clothing type.

4 Results

5 Discussion

5.1 Limitations

5.2 Future Work

5.3 Conclusions

References

[1] Neil Lawrence. Probabilistic Non-linear Principal Component Analysis with Gaussian Process Latent Variable Models. *Journal of Machine Learning Research*, 6:1783–1816, November 2005.

[2] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An Imperative Style, High-Performance Deep Learning Library, 2019. URL https://arxiv.org/abs/1912.01703.

[3] Carl Edward Rasmussen and Christopher K.I. Williams. *Gaussian Processes for Machine Learning*. The MIT Press, 2006. URL https://gaussianprocess.org/gpml/.

[4] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms, 2017.